Astrophysics/Physics Honours Research Skills Class 5: How to use AI tools

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Welcome to your graduate research journey. Artificial Intelligence is no longer a novelty; it is a fundamental instrument in the modern astronomer's toolkit. However, simply "chatting" with an AI is the absolute baseline. These notes bypass the obvious and focus on novel, actionable, and professional workflows to accelerate your research without compromising your scientific integrity.

Crucially, remember that Large Language Models (LLMs) are pattern-completion engines, not reasoning machines. They have no ground-truth model of physics. They are fundamentally different from the classical Machine Learning (Random Forests, CNNs) you will use to classify galaxies or detect transients.

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1. AI Basics: The Landscape and Interfaces

1.1 The 2026 Provider Ecosystem

You should be comfortable switching between providers based on the task.

Provider / Model (Early 2026)	Best Use Case in Astrophysics
Anthropic (Claude 4)	Deep reasoning, massive context windows (reading 50+ PDFs), agentic coding.
OpenAI (GPT-5 / o-series)	Complex mathematical reasoning, advanced data analysis, versatile API tooling.
Google (Gemini 3)	Multimodal tasks (analyzing plots/images), Google Workspace integration.
Meta / DeepSeek (Llama 4 / V4)	Open-weight models. Run locally for absolute privacy with embargoed data.

1.2 Interfaces: Chat Apps vs. APIs and Agents

- **The Baseline:** Using the web interfaces of frontier models for general brainstorming and basic text generation.
- **The Pro-Move:** Transitioning from web chats to purpose-built agentic tools. For coding, AI-native editors like **Windsurf** or **Cursor** index your entire repository, allowing the AI to autonomously execute multi-step tasks across dozens of files. For data processing, use Python APIs to batch-process thousands of catalog entries programmatically.

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2. AI Ethics, Data Privacy, and Scientific Integrity

2.1 Data Privacy and Embargoed Science

- **The Baseline:** Assuming all AI chat interfaces are private and secure.
- **The Pro-Move:** Treating consumer AI like a public billboard. **Never** upload unreleased JWST, ALMA, or proprietary collaboration data to public consumer AI models (like ChatGPT's free tier) where it might be ingested for future model training. If you are working with embargoed telescope data or pre-publication results, use enterprise API endpoints (which typically have strict zero-data-retention policies) or run open-weight models (like Llama 4) locally on your own hardware.

2.2 Hallucinations and the "Human-in-the-Loop" Mandate

- **The Baseline:** Knowing that AI occasionally makes mistakes or typos.
- **The Pro-Move:** Recognizing that LLMs are probabilistic text generators with *no underlying physical world model*. They will confidently hallucinate fake papers, invent non-existent Python library functions, and derive mathematically flawed proofs. You must employ a strict "Human-in-the-Loop" (HIL) workflow. You bear 100% of the responsibility for every factual inaccuracy or hallucinated error in your code or manuscript.

2.3 Transparency, Plagiarism, and Journal Disclosures

- **The Baseline:** Hiding your AI usage out of fear of academic penalty, or copying AI text verbatim.
- **The Pro-Move:** Radical transparency. Passing off AI-generated scientific narratives or novel hypotheses as your own original thought is a breach of academic integrity. Major astronomical journals (ApJ, MNRAS, A&A) and university integrity boards now explicitly require AI disclosure in your methodology or acknowledgments.
    - *Template:* "The authors acknowledge the use of [Model] to support[specific task, e.g., drafting the introduction/refactoring Python scripts]. All AI-generated suggestions were reviewed, verified, and approved by the authors."

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3. Using AI to Get Work Done (vs. Shortcut Learning)

3.1 The Learning vs. Productivity Distinction

The single most important principle of graduate school: **AI should amplify your thinking, not replace it.**

✅ Augmenting (Do This)	❌ Outsourcing (Don't Do This)
Using AI to explain a cryptic compiler error.	Having AI write your entire data pipeline from scratch.
Using AI to find logical gaps in your derivation.	Asking AI to derive the equation for you.
Using AI to suggest an outline for your thesis chapter.	Submitting AI-generated prose as your own writing.

- **The Golden Rule:** If you could not explain, debug, or reproduce the AI's output *without* the AI, you have outsourced your cognition. This will critically stunt your development as an independent scientist.
- **The Socratic Co-Pilot:** Use AI for "Rubber Duck Debugging." Prompt: *"I am trying to derive the radiative transfer equation for this cloud geometry. Here is my math. Do not solve it, but tell me where my assumptions are failing."*

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4. Smart Prompting: Making the AI Do What You Want

4.1 Persona, Few-Shot Formatting, and Constraints

- **The Baseline:** "Make a table of these stars."
- **The Pro-Move:** Assigning a strict persona and using "few-shot" examples. Note: giving an AI a persona (e.g., "Act as an astrophysicist") helps with *tone and formatting*, but research shows it does little to improve factual accuracy. For accuracy, you need examples.
    - *Prompt: "Format these stellar masses into a LaTeX AASTeX631 table suitable for ApJ. Here is exactly how one row should look: [Insert Example]. Now process the rest."*

4.2 Chain-of-Thought for Physics Workflows

- **The Pro-Move:** Force the AI to think step-by-step before writing code. Prompt: *"Before writing the Python script to extract transit light curves from this FITS file, outline the physical assumptions, the background subtraction method, and the mathematical steps you will take."* Catch physics errors in plain English before they hide in Python syntax.

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5. AI to Communicate (Emails, Reports, Thesis)

5.1 Reverse-Engineering Grant Proposals

- **The Baseline:** Asking AI to "write a telescope proposal." (It will be rejected).
- **The Pro-Move:** Reverse-engineering success. Feed a successful, funded telescope time proposal into an LLM with a massive context window. Ask it to extract the structural template, the rhetorical strategies used to justify the science case, and the tone. Use that extracted template as the skeleton for your original science case.

5.2 Overcoming "Blank Page" Syndrome

- **The Pro-Move:** Dump your messy, bulleted lab notes and plot descriptions into an LLM and ask it to draft a rough "Methods" section. The output will be dry and generic, but editing a mediocre draft is psychologically much easier than staring at a blank LaTeX document. You maintain your scientific voice by heavily editing the AI's scaffolding.

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6. AI for Navigating and Understanding the Literature

6.1 Discovery and the Citation Graph

Traditional keyword searches in the Astrophysics Data System (ADS) miss semantically related work. Use AI to bridge the gap, but be aware that many AI tools rely on Semantic Scholar, which may miss niche astronomy papers. Use these to *complement* ADS, not replace it:

Tool	Best Use Case
Elicit / Consensus	Asking natural-language questions to extract evidence across dozens of papers.
Litmaps / Research Rabbit	Visualizing citation networks to find foundational papers in a new sub-field.
Scite.ai	Checking if a specific paper's methodology has been supported or contrasted by downstream citations.

6.2 Massive Context Windows & Parameter Extraction

- **The Pro-Move:** Drop 50 peer-reviewed PDFs directly into Claude 4 or Gemini 3 and prompt: *"Read these PDFs and extract the reported exoplanet radii, error margins, and the specific spectrograph used into a CSV file."* This grounds the AI entirely in the literature you provide, preventing hallucinations.

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7. AI Coding (Shell Scripting, Python, C, etc.)

7.1 What AI Does Well vs. What it Fails At

- **Where it excels:** Generating boilerplate, writing `pytest` suites, explaining legacy Fortran/C code, and writing arcane SLURM bash scripts for High-Performance Computing (HPC) clusters.
- **Where it fails:** Complex refactoring across large, multi-file codebases. AI struggles to maintain behavior while restructuring deep architectural logic.
- **The Pro-Move:** Delegating the boilerplate. Write the core mathematical functions yourself (where the physics lives), then prompt the AI: *"Write a comprehensive `pytest` suite for this function, including edge cases for negative fluxes and NaN values."*

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8. AI for Project Management

8.1 The Friday Weekly Summary

- **The Pro-Move:** A PhD involves hundreds of micro-tasks. At the end of every week, paste your messy daily notes into an LLM and ask it to generate:
  1. A structured summary of progress.
  2. Outstanding tasks, prioritized.
  3. Blockers.
  4. A bulleted list of questions for your next supervisor meeting.
     This creates an invaluable running log for your thesis writing.

8.2 Micro-Tasking the PhD Overwhelm

- **The Pro-Move:** "Write Chapter 2" is a paralyzing task. Feed your chapter outline to an AI and ask it to break it down into 2-hour, actionable "Pomodoro" blocks. *"Draft 3 paragraphs on the history of 21cm cosmology instrumentation"* is an

actionable task you can actually complete before lunch.

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9. Discussion Points for Class

1. **The Line:** Where is the line between using AI to learn and using AI to bypass learning? Does it shift between your coursework and your actual research?
2. **Data Privacy:** What would you *never* paste into ChatGPT? Where is your personal data-sensitivity boundary?
3. **The LSST Era:** Vera Rubin's LSST will generate 10 million alerts per night. How does the rise of LLMs and agentic AI change how we might triage this data compared to classical ML?
4. **Institutional Policy:** Have you read our university's specific policy on AI use in theses and publications? What does it actually say?
